

Existing Solutions and Comparison

1. Method

Six existing systems were reviewed against eight criteria derived from the capabilities Itqan is designed to provide. Each cell in the comparison table is supported by a statement published by the vendor on its own website, help center, app-store listing or privacy policy. The sources are listed in Section 4.

Two rules were applied throughout the review:

- A capability is recorded as **"not documented"** rather than as absent when no source states either way. Silence in a feature list is not evidence of absence.
- Vendor marketing claims are recorded as claims. Where a vendor advertises a capability but publishes no evaluation of it, the table states this.

The systems were identified by searching the Apple App Store, Google Play and the web in both English and Arabic. This method has a limitation: a system that is not marketed through these channels may not have been identified.

2. Comparison of Existing Systems

System	Listens to the user	Word-level mistakes	Tajweed-rule detection	Error shown in the text	Help when the reciter pauses	Teacher / class monitoring	Scope
Tarteel	Yes, live, server-side	Yes — missed, incorrect and skipped words, and tashkeel	No — the vendor states the AI is "not trained to recognize tajweed mistakes yet"	Yes — the word is highlighted and can be tapped to compare the recitation with the correct text	Yes — detects that the reciter is stuck; hint controls reveal the next word or the rest of the ayah	Not documented; there is no institutional plan, and Family Plan members cannot view each other's progress	Full Quran (Hafs only)
Tilawa.ai	Yes	Yes — "word-by-word mistakes and skipped words"	Claims "over 16 tajweed rules" and makharij feedback; no evaluation published	Not documented how errors are anchored to the text	Not documented	Not documented	Full Quran
Thurayya	Yes	Corrects "every ayah on the spot"	Claims named-rule correction at word level	Claimed at word level	Not documented	Progress tracking for parents	Juz Amma only
Tasmi	Yes	Yes — "matched, missed, and wrong words"	No — the vendor states "Tajweed is for your teacher"	Yes — matched, missed and wrong words are shown	Not documented	Yes — teachers and parents assign passages and confirm practice	Not stated
Qurani.ai (QRC API)	Yes, real-time	Yes — returns <code>correct_words</code> and <code>skipped_words</code>	Yes — returns <code>tajweed_mistakes</code> separately from <code>letter_mistakes</code>	Returns chapter, verse and word indices	Not applicable (API)	No — the product is an API with no end-user application	Not stated
TajweedMate	Records, then analyzes	Not documented	Claims Huruf, Madd, Ith'har, Ikhfaa, Iqlaab and Idghaam	Not documented	Not documented	Not documented	Not stated
Itqan (proposed)	Yes	Yes	Yes (AI-Mujawwid)	Yes	Yes — Smart Prompting	Yes (AI-Mu'allim)	Juz Amma, 37 Surahs

3. Capabilities Already Provided by Existing Systems

The following capabilities are already provided by one or more of the reviewed systems. Itqan includes them because they are necessary to the proposed solution, not because they are new:

- **Real-time word-level mistake detection.** Tarteel has provided this capability since 2018 and advertises the detection of "missed, incorrect and skipped words in your recitation."
- **Indicating the error within the Quranic text.** Tarteel highlights the erroneous word and allows the user to tap it in order to see "exactly what you recited versus the correct Ayah."
- **Corrective audio.** Tarteel instructs the user to "Tap the reciter button next to it to hear the correct recitation," and Tilawa provides word-by-word reference audio.
- **Assisting a reciter who pauses.** Tarteel states that "if you get stuck, our A.I will be sure to let you know," and its Testing mode provides reveal controls in which "each reveal counts as a mistake."
- **Tajweed feedback as a capability.** Tilawa, Thurayya, TajweedMate and the Qurani.ai API each offer it.
- **Teacher and class monitoring.** Tasmi allows teachers and parents to "assign passages and confirm practice happened."
- **A Juz Amma scope.** Thurayya is scoped to Juz Amma and is directed at children.
- **Streaks, progress tracking and session scores.** These are present across most of the reviewed systems; Tarteel tracks memorization "all the way to the individual word level."

Accordingly, Itqan is not presented as the first or the only AI-based Quran recitation system, and none of the capabilities listed above is claimed as a novel contribution of this project.

Two further observations emerged from the review. First, none of the reviewed systems publishes an accuracy figure for recitation *error detection*; the only public figure identified is a 4% word-error rate for Arabic *transcription*, which measures a different property. Second, Tarteel's opt-out from AI training and its in-app deletion of recordings are both paid features, and its privacy policy states no retention period for recitation audio.

4. Sources

Tarteel — tarteel.ai · tarteel.ai/pricing · tarteel.ai/privacy · support.tarteel.ai (articles 12414457 tajweed, 12414416 and 16557456 prompting and hints, 14423894 corrective audio, 12414504 AI-training opt-out, 12414461 deleting recordings, 12414428 Family Plan privacy, 12414396 riwayat, 12414407 memorization tracker)

Tilawa.ai — tilawa.ai/en · tilawa.ai/en/privacy · tilawa.ai/en/terms · App Store id6760923881

Thurayya — thealphazed.com/thurayya · App Store id1571034432

Tasmi — tasmi.app

Qurani.ai — qurani.ai/en/docs/2-advanced-tools/qrc

TajweedMate — tajweedmate.com

Transcription accuracy figure — nvidia.com/en-gb/case-studies/automating-real-time-arabic-speech-recognition (undated)